

BOGOTOL, Russian Federation

WMO: 295530

Lat: 56.2291N

Lon: 89.5535E

Elev: 291

StdP: 97.88

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.1	(c) -32.8	(d) -39.1	(e) 0.1	(f) -35.9	(g) -36.1	(h) 0.1	(i) -32.7	(j) 9.3	(k) -7.9	(l) 8.2	(m) -10.5	(n) 0.9	(o) 50	(p) 0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.3	(c) 28.0	(d) 19.0	(e) 26.3	(f) 18.1	(g) 24.6	(h) 17.4	(i) 20.5	(j) 25.5	(k) 19.4	(l) 24.3	(m) 18.4	(n) 23.2	(o) 2.3	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.1	22.9	17.6	13.0	22.0	16.5	12.2	21.1	60.6	25.9	57.0	24.2	53.4	23.2	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.3	(b) 6.4	(c) 5.6	(e) -37.9	(f) 30.7	(g) 5.0	(h) 1.5	(i) -41.4	(j) 31.8	(k) -44.4	(l) 32.6	(m) -47.2	(n) 33.5	(o) -50.8	(p) 34.5		
(a) 7.3	(b) 6.4	(c) 5.6	(e) -38.0	(f) 22.1	(g) 4.9	(h) 1.5	(i) -41.5	(j) 23.1	(k) -44.4	(l) 24.0	(m) -47.2	(n) 24.9	(o) -50.8	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-17.0	-13.9	-6.2	2.3	9.3	16.5	17.9	15.3	9.0	1.4	-8.2	-14.5		
	DBStd	13.66	8.70	7.63	6.16	5.59	5.63	3.92	3.15	3.57	4.53	5.60	7.56	8.18		
	HDD10.0	3989	838	671	502	236	84	4	0	4	72	271	547	759		
	HDD18.3	6376	1097	904	761	481	284	76	47	107	279	526	797	1018		
	CDD10.0	727	0	0	0	5	63	201	243	167	43	4	0	0		
	CDD18.3	71	0	0	0	0	4	22	32	12	1	0	0	0		
	CDH23.3	641	0	0	0	1	55	236	238	97	14	0	0	0		
	CDH26.7	113	0	0	0	0	12	44	37	17	2	0	0	0		
(14) Wind	WSAvg	2.4	2.6	2.5	2.8	2.7	2.5	1.9	1.6	1.7	2.1	2.6	2.7	2.8		
(15) Precipitation	PrecAvg	521	24	20	24	31	43	65	58	70	50	53	46	38		
	PrecMax	688	39	41	61	56	112	123	105	166	101	87	69	68		
	PrecMin	405	4	6	3	6	15	13	17	29	5	22	25	10		
	PrecStd	68	9	9	13	12	23	25	25	28	23	15	14	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.8	8.5	19.5	28.3	30.1	29.5	28.9	25.5	17.2	8.2	1.0		
		MCWB	-1.4	-0.8	3.7	10.1	17.0	19.4	20.5	20.7	16.4	9.9	5.6	-0.5		
	2%	DB	-2.0	-0.3	5.2	16.1	24.5	27.9	27.7	25.9	21.3	14.1	5.1	-0.8		
		MCWB	-3.6	-2.1	1.4	8.5	14.0	18.1	19.9	18.9	14.4	8.7	3.0	-2.2		
	5%	DB	-3.8	-1.8	3.5	13.6	21.8	26.0	26.0	23.8	18.8	11.6	2.7	-2.4		
		MCWB	-5.1	-3.4	0.5	7.3	12.5	17.3	19.0	17.5	12.8	7.2	0.9	-3.5		
	10%	DB	-5.7	-3.7	1.9	11.0	19.0	24.2	24.3	22.0	16.4	9.2	0.7	-4.5		
		MCWB	-6.8	-5.3	-0.7	5.7	11.5	16.5	18.1	16.5	11.6	5.9	-0.7	-5.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	-0.5	3.8	10.9	17.4	20.9	22.1	21.8	16.8	11.1	5.9	-0.3		
		MCDB	0.2	1.3	7.7	17.6	27.9	27.6	27.5	26.4	24.4	15.6	8.4	0.5		
	2%	WB	-3.4	-1.9	2.1	9.3	14.9	19.4	20.8	19.9	14.9	9.0	3.2	-2.0		
		MCDB	-2.1	-0.5	4.8	14.9	22.3	25.4	25.7	24.1	20.3	13.5	5.1	-0.9		
	5%	WB	-5.0	-3.4	0.6	7.7	13.4	18.3	19.9	18.5	13.6	7.6	1.0	-3.5		
		MCDB	-3.8	-1.8	3.0	13.2	20.0	24.4	24.6	22.7	17.5	11.0	2.4	-2.3		
	10%	WB	-6.8	-5.2	-0.6	5.8	12.0	17.3	18.9	17.2	12.3	6.0	-0.6	-5.5		
		MCDB	-5.6	-3.7	1.9	10.5	18.4	22.6	23.2	21.1	15.7	9.2	0.7	-4.5		
(35) Mean Daily Temperature Range	MDBR		6.4	7.1	8.0	8.9	10.1	10.3	9.3	9.3	8.6	6.2	5.7	6.4		
	5% DB	MCDBR	6.3	6.6	8.8	12.7	14.0	14.2	12.0	12.4	13.1	10.2	5.8	6.7		
		MCWBR	5.6	5.9	6.4	7.4	7.1	6.4	5.5	6.3	7.0	6.1	4.4	6.2		
	5% WB	MCDBR	6.4	6.6	8.1	11.9	12.4	12.0	10.5	10.9	11.6	9.6	5.7	6.6		
		MCWBR	5.8	6.0	6.3	7.4	6.9	5.9	5.5	6.0	6.7	6.0	4.6	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.365	0.394	0.405	0.403	0.427	0.406	0.369	0.348	0.314	0.284		
	taud		2.001	1.980	1.923	2.082	2.202	2.297	2.229	2.267	2.318	2.220	2.028	1.945		
	Ebn at Noon		568	705	765	814	832	840	808	800	777	677	517	444		
	Edh at Noon		66	102	139	139	131	121	127	115	95	80	64	54		
(44) All-Sky Solar Radiation	RadAvg		0.77	1.74	3.08	4.31	4.94	5.86	5.30	4.23	2.76	1.42	0.69	0.46		
	RadStd		0.09	0.15	0.24	0.32	0.55	0.54	0.37	0.34	0.35	0.13	0.10	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page